

**memory-bound
(bandwidth)**

**compute-bound
(FLOPs)**

**roofline
ridge ~295**

decode @ low batch



prefill @ 9.2K prompt



HBM weight-stream per token:
~1 FLOP/byte at batch-1 (Ch8)

$2 \cdot N \cdot L$ over HBM weight stream:
high intensity

lower →

→ higher

arithmetic intensity (FLOP/byte) →

batch size ↑ · context ↑ · KV precision ↓ · kernel/hardware → all move a point across the ridge